Homework 2: Panel & RDD

Start Assignment

- Due Sunday by 11:59pm
- Points 25
- Submitting a file upload

Homework 2: Data Analysis and Statistical Decision Making
May 2026

The submission deadline is **Sunday May 24** at midnight. There will be a deduction of 10% for every day of late submission, up to 3 days. Please submit an .Rmd or .R file **and** a PDF with all your answers, tables and figures. Make sure to number the questions. Pay attention to formatting of figures and tables: include titles, legible labels, and standard errors in all regression tables. You may use AI tools to assist you in coding, but you must be able to explain every line of code and every interpretation in your submission. You may be asked to do so in office hours/recitation.

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Homework_2_QM3-1.pdf (https://canvas.ucsd.edu/courses/74851/files/18293991?wrap=1) (https://canvas.ucsd.edu/courses/74851/files/18293991/download?download_frd=1)

Part I: Panel and Two-Way Fixed-Effects (12 Points)

Africa_GDP.Rda (https://canvas.ucsd.edu/courses/74851/files/18293847?wrap=1) (https://canvas.ucsd.edu/courses/74851/files/18293847/download?download_frd=1)

The file **Africa_gdp** gives a measure of political liberties for 46 African countries from 1985 to 1998 (the measure is scaled from 0 (worst) to 4 (best)). We want to understand the relationship between income and good governance.

Please go to the WDI data and download per-capita GDP in constant US dollars for each of these countries for the period covered by the governance data, and join it to the main database.

1. Generate a numeric country identifier and a set of year dummies. Estimate the pooled OLS, and within regression for the relationship between governance (RHS) and income (LHS), controlling for year dummies where possible. Present results in a single table.
2. The variable bigimp is a dummy equal to one for the year in which each country that improved saw its largest improvement in political liberties. We want to understand whether income increases *before* or *after* governance improves. First, run a two-way fixed effects regression (using only year and country dummies) with per capita GDP as the outcome. Then, use an event study framework to examine the leads and lags of large governance improvements on GDP. You may use a package in R (such as

eventstudies, did, or bacondecomp) or construct it manually: generate a variable leadlag that counts from −5 to +5, where −5 means 5 years before the big improvement and +5 means 5 years after. Plot the residuals from the regression against the range of leadlag; use the code from lab 5 as guidance.

3. Interpret the results from (1) and the figure in (2) in a few concise sentences. Are countries that are richer on average better governed? Should we expect to see political improvements in African countries with rapidly expanding economies? Give some intuition for what you observe.
4. The results so far pertain to the average country in Africa, not the average person. Acquire the additional data needed to make the analysis correct for the representative person rather than the representative country. Use one table, one figure, and one paragraph to describe how your answers change and what this means.

Part II: Regression Discontinuity Design (13 Points)

grade5.dta (https://canvas.ucsd.edu/courses/74851/files/18293848?wrap=1)
(https://canvas.ucsd.edu/courses/74851/files/18293848/download?download_frd=1)

rdd_paper.pdf (https://canvas.ucsd.edu/courses/74851/files/18293857?wrap=1)
(https://canvas.ucsd.edu/courses/74851/files/18293857/download?download_frd=1)

For this part, you will use the framework from the paper: Angrist, J. D., and V. Lavy (1999). "Using Maimonides' Rule to Estimate the Effect of Class Size on Scholastic Achievement." *Quarterly Journal of Economics*, 114(2): 533–575.

For the analysis, you will use the dataset grade5.dta, originally from Raj Chetty's course "Big Data to Solve Economic and Social Problems." It contains school-level data for fifth graders. Key variables: school_enrollment (running variable), classize, avgmath, avgverb, and disadvantaged (share of students from disadvantaged backgrounds).

1. Read the intro of the Angrist & Lavy (1999) paper. Explain what endogeneity problem arises if we estimate the effect of class size on test scores using a simple OLS regression. What variables might be omitted? Would the regression over- or underestimate the true effect? Justify your answer.
2. Plot a histogram of school_enrollment. Do you think parents take the Maimonides Rule into account when choosing a school? Why or why not?
3. Plot in a way you find appropriate the relationship between classize and math scores, and between classize and verbal scores. Include the Maimonides Rule cutoff in your plot. What do you observe? Does the evidence justify the preference for smaller classes?
4. Estimate the RD effect of crossing the enrollment cutoff of 40 on avgmath and avgverb, restricting to schools with fewer than 80 students enrolled.
   1. First, implement the estimator manually: run a local regression on each side of the cutoff using a bandwidth of your choice. Report the coefficient, bandwidth, and effective N on each side. Justify your answer.

    2. Now, re-estimate using rdrobust (with default settings). Report the coefficient, bandwidth, effective N on each side, and 95% confidence interval. Compare the previous answer and discuss the differences.

5. Choose a falsification test and run it — for example, a density test on the running variable, a smoothness test using a pre-determined covariate as the outcome, or a placebo cutoff point. Briefly explain why you chose it. Present your results and discuss what a failure would imply for your estimates.